

Trainings ▾

Install

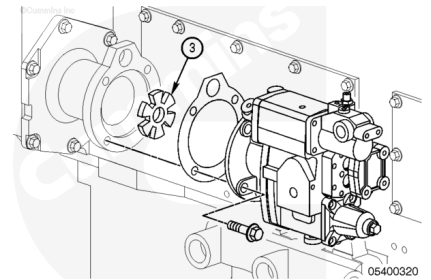
with Mechanically Actuated Injector

All QSK19 engines use a light green fuel pump drive coupling.

Install the fuel pump drive coupling (3), gasket, fuel pump, and four capscrews.

Tighten the capscrews.

Torque Value: 45 n•m [33 ft-lb]



Check for the proper installation of the o-rings in both locations.

Install the fuel pump inlet hose and the fuel supply hose.

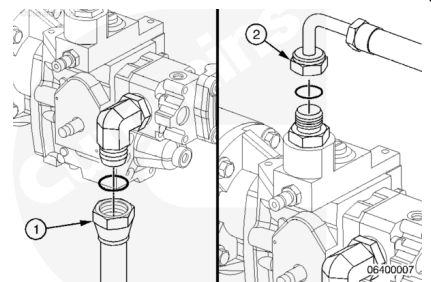
Tighten the hoses.

Torque Value:

Fuel Pump Inlet Hose 88 n•m [65 ft-lb]

Torque Value:

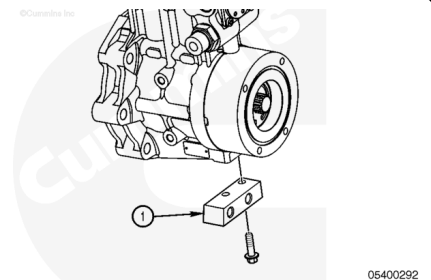
Fuel Supply Hose 60 n•m [44 ft-lb]



with Electronically Actuated Injector

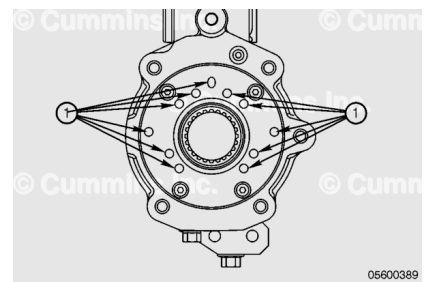
Install the fuel pump support bracket (1) to the fuel pump with two capscrews.

Torque Value: 113 n•m [83 ft-lb]



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

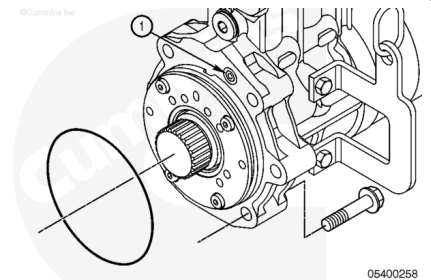


On a flat level surface, place the pump on end, with the gerotor-side down.

Pour 104 ml [3.5 oz] of fresh, clean lubricating oil into one of the eleven available face drillings (1).

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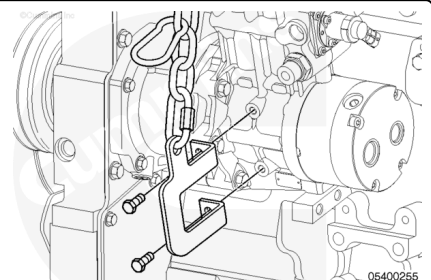


Use clean engine oil to lubricate both new o-rings.

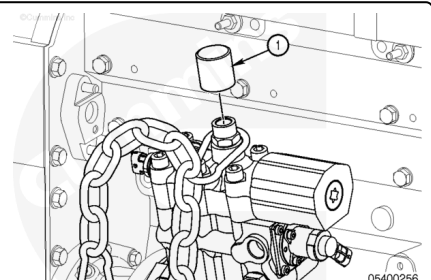
Install the smaller o-ring (1) to seal the oil drilling from the fuel pump drive to the fuel pump.

Install the larger o-ring to seal the fuel pump drive to the fuel pump.

Install the bracket onto the fuel pump using two M10 capscrews.

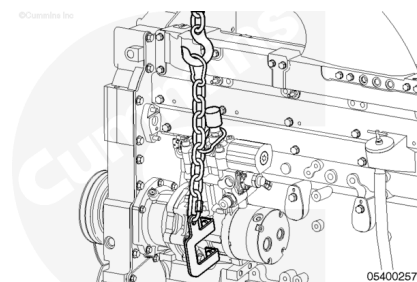


Place the bottom large lifting connector over the fitting boss on the top of the fuel pump. Screw the stop (1) into the fitting on top of the fuel pump until it is hand-tight.



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Attach the top large lifting connector to a suitable lifting device (capable of lifting at least 227 kg [500 lb]).

Raise the lifting device until there is no slack in the chain.

Install the fuel pump onto the engine. The lifting tool has a provision for installing a breaker bar to aid in aligning the fuel pump.

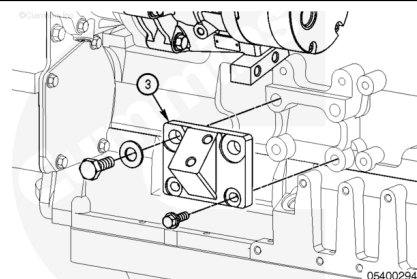
Secure the fuel pump with five capscrews.

Torque Value: 113 N•m [83 ft-lb]

Remove the fuel pump installation and removal service tool.

Install the fuel pump support bracket (3) to the engine. Hand tighten the capscrews.

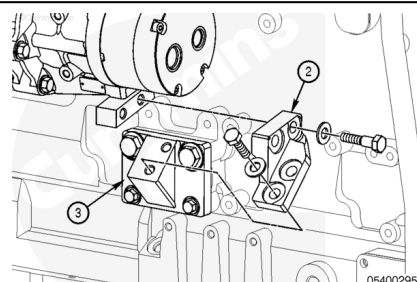
The fuel pump support bracket holes are enlarged to aid in alignment. Final tightening will occur when all the brackets are installed and aligned.



Secure the fuel pump support bracket (2) to the support bracket (3) and the fuel pump with capscrews. Tighten the capscrews finger-tight.

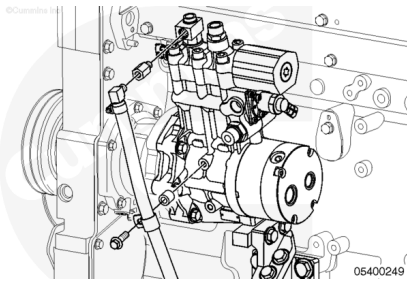
The bracket holes are enlarged to aid in alignment. Final tightening will occur when all the brackets are installed and aligned.

With all the capscrews in place and finger tight, tighten all the capscrews.



Thread	Socket Size	N•m	[ft-lb]
3/8-16	9/16	40	30
1/2-13	3/4	130	95
5/8-11	15/16	255	190

Install a clamp on the injector supply line vent to the fuel pump and on the engine block where the injector supply line attaches.



Connect the wiring harness to the fuel pump actuator and fuel pump pressure sensor.

